

CS536 Science of Programming
Fall 2025
Assignment 2 Sample Solution Sketches

1. Recall that syntactic equality logically implies semantic equality.

		$\sigma[u \mapsto \alpha][v \mapsto \beta] = \sigma[v \mapsto \beta][u \mapsto \alpha]$?	$\sigma[u \mapsto \alpha][v \mapsto \beta] \equiv \sigma[v \mapsto \beta][u \mapsto \alpha]$?
$u \equiv v$	$\alpha = \beta$	Yes, because they are syntactically equal.	Yes, texts are textually the same on both hand-sides: we are updating the same variable with the same value twice.
$u \equiv v$	$\alpha \neq \beta$	No. On the left-hand-side, the variable $u \equiv v$ is bound with β ; and on the right-hand-side, the variable $u \equiv v$ is bound with α .	No, because they are not semantically equal.
$u \not\equiv v$	$\alpha = \beta$	Yes. On both hand-sides, both variables u and v are bound with α (or β , since they are of the same value).	No, they are different procedures since u and v are different variables.
$u \not\equiv v$	$\alpha \neq \beta$	Yes. On both hand-sides, variable u is bound with α and variable v is bound with β .	No, they are different procedures since u and v are different variables.

2. Recall that, a state σ is no longer σ after updating.

- (a) $\sigma[x \mapsto \sigma(y)][y \mapsto \sigma(x)] = \sigma[x \mapsto 4][y \mapsto \sigma(x)] = \sigma[x \mapsto 4][y \mapsto 2] = \{x = 4, y = 2\}$.
- (b) $\sigma[x \mapsto \sigma(y)][x \mapsto \sigma[x \mapsto \sigma(y)](x - y)] = \sigma[x \mapsto 4][x \mapsto \sigma[x \mapsto 4](x - y)] = \sigma[x \mapsto 2][x \mapsto \{x = 4, y = 4\}(x - y)] = \sigma[x \mapsto 4][x \mapsto 0] = \{x = 0, y = 4\}$
- (c) $\gamma = \sigma[x \mapsto 3][y \mapsto \tau(x) * 4] = \sigma[x \mapsto 3][y \mapsto \sigma[x \mapsto 3](x) * 4] = \sigma[x \mapsto 3][y \mapsto 12] = \{x = 3, y = 12\}$.

3. (a) True. $\{x = 4, y = 2\}(x < 3 \rightarrow y > 2) \Leftrightarrow (F \rightarrow F) \Leftrightarrow T$.
- (b) True. We can show that the state satisfies $x > 1$ and also satisfies $\exists x. 0 \leq x < 4 \wedge b[x] < 3$. The satisfaction of $x > 1$ is obvious. To show the satisfaction of $\exists x. 0 \leq x < 4 \wedge b[x] < 3$, we can find a witness $x = 0$.
- (c) True. We can find $y = 2$ as a witness because $\sigma(b[0]) = \sigma(b[1]) = 2$.
4. (a) $x := 0; i := 0; \text{ while } i < \text{size}(b) \text{ do } x := x + b[i]; i := i + 1 \text{ od}$
- (b) **while** $x \neq 1$ **do** **if** $x \% 2 = 0$ **then** $x := x/2$ **else** $x := x + 1$ **fi od**
- (c) $m := 8; p := 1; y := 1; m := m + 1; \text{ while } m < 20 \text{ do } p := p * y; y := y + 1; m := m + 1 \text{ od}$

5. (a)

$$\begin{aligned}
& \langle \text{if } x < 2 \text{ then } x := y + 1, w := x + 2 \text{ fi}, \sigma \rangle \\
& \rightarrow \langle \text{skip}, \sigma \rangle && \text{because } \sigma(x) \geq 2 \\
& \rightarrow \langle E, \sigma \rangle
\end{aligned}$$

(b)

$$\begin{aligned}
& \langle x := y + 1; y := x + 1, \sigma \rangle \\
& \rightarrow \langle y := x + 1, \sigma[x \mapsto \sigma(y) + 1] \rangle \\
& \rightarrow \langle E, \sigma[x \mapsto \sigma(y) + 1][y \mapsto \sigma[x \mapsto \sigma(y) + 1](x) + 1] \rangle \\
& = \langle E, \sigma[x \mapsto \sigma(y) + 1][y \mapsto \sigma(y) + 2] \rangle
\end{aligned}$$

(c)

$$\begin{aligned}
& \langle W, \sigma \rangle \\
& \rightarrow \langle S; W, \sigma \rangle && \text{because } \sigma(x) > \sigma(y) \\
& \rightarrow \langle y := -2 * x; W, \sigma \rangle && \text{because } \sigma(x) \leq 0 \\
& \rightarrow \langle W, \sigma[y \mapsto -2 * \sigma(x)] \rangle \\
& \rightarrow \langle E, \sigma[y \mapsto -2 * \sigma(x)] \rangle && \text{because } \sigma(x) \leq 0 \text{ and } \sigma[y \mapsto -2 * \sigma(x)](y) \geq 0
\end{aligned}$$

6. (a)

$$\begin{aligned}
& M(S, \sigma_1) \\
& = M(p := p \wedge (x > 1), \sigma_1[x \mapsto \sigma_1(x) + 1]) \\
& = M(E, \sigma_1[x \mapsto \sigma_1(x) + 1][p \mapsto \sigma_1[x \mapsto \sigma_1(x) + 1](p \wedge (x > 1))]) \\
& = \{\sigma_1[x \mapsto \sigma_1(x) + 1][p \mapsto \sigma_1[x \mapsto \sigma_1(x) + 1](p \wedge (x > 1))]\} \\
& = \{\sigma_1[x \mapsto \sigma_1(x) + 1][p \mapsto \sigma_1(p) \wedge (\sigma_1(x) > 1)]\}
\end{aligned}$$

(b) $M(W, \sigma_2) = \{\sigma_2\}$, because $\sigma_2(x < 2) = F$.

(c)

$$\begin{aligned}
& M(W, \sigma_3) \\
& = M(x := x + 1; p := p \wedge (x > 1); W, \sigma_3) && \text{because } \sigma_3(x < 2) = T \\
& = M(p := p \wedge (x > 1); W, \sigma_3[x \mapsto 2]) \\
& = M(W, \sigma_3[x \mapsto 2][p \mapsto \sigma_3[x \mapsto 2](p \wedge (x > 1))]) \\
& = M(W, \sigma_3[x \mapsto 2][p \mapsto T]) && \sigma_3(p) = T \text{ and } x \text{ is bound with } 2 \\
& = \{\sigma_3[x \mapsto 2][p \mapsto T]\} && \sigma_3[x \mapsto 2][p \mapsto T](x < 2) = F
\end{aligned}$$

7. (a) $M(S, \sigma_1) = M(x := y/x, \sigma_1) = M(E, \sigma_1[x \mapsto \sigma_1(y/x)]) = \{\sigma_1[x \mapsto 0]\}$

(b)

$$\begin{aligned}
& M(W, \sigma_2) & \sigma_2 = \{x = 2, y = 2, b = (0, 1, 2)\} \\
& = M(S; W, \sigma) & \sigma_2(x > 0) = T \\
& = M(x := x - 1; y := b[y]; W, \sigma_2) & \sigma_2(x < y) = F \\
& = M(W, \sigma_2[x \mapsto 1][y \mapsto 2]) \\
& = M(S; W, \sigma_2[x \mapsto 1][y \mapsto 2]) & \sigma_2[x \mapsto 1][y \mapsto 2](x > 0) = T \\
& = M(x := y/x; W, \sigma_2[x \mapsto 1][y \mapsto 2]) & \sigma_2[x \mapsto 1][y \mapsto 2](x < y) = T \\
& = M(W, \{x = 2, y = 2, b = (0, 1, 2)\}) \\
& = \{\perp_d\} & \text{the same configuration appears again}
\end{aligned}$$

(c)

$$\begin{aligned}
& M(W, \sigma_3) \\
& = M(W, M(S, \sigma_3)) & \text{because } \sigma_3(x > 0) = T \\
& = M(W, M(x := x - 1; y := b[y], \sigma_3)) & \sigma_3(x < y) = F \\
& = M(W, \{x = 7, y = 0, b = (4, 2, 0)\}) & \text{after the first iteration} \\
& = M(W, M(S, \{x = 7, y = 0, b = (4, 2, 0)\})) & \text{because } x > 0 \text{ is evaluated to T} \\
& = M(W, \{x = 6, y = 4, b = (4, 2, 0)\}) & \text{after another iteration} \\
& = M(W, M(S, \{x = 6, y = 4, b = (4, 2, 0)\})) & \text{because } x > 0 \text{ is evaluated to T} \\
& = \{\perp_e\} & \text{array index out of bounds in b[y]}
\end{aligned}$$

8. (a) $\sigma \models \exists x \in S.p$ means for **this** state σ and for **some** $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \models p$.
- (b) $\sigma \models \forall x \in S.p$ means for **this** state σ and for **all** $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \models p$.
- (c) $\sigma \not\models \exists x \in S.p$ means for **this** state σ and for **all** $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \not\models p$.
- (d) $\sigma \not\models \forall x \in S.p$ means for **this** state σ and for **some** $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \not\models p$.
- (e) $\models \exists x \in S.p$ means for **all** state σ , it is the case that $\sigma \models \exists x \in S.p$.
- (f) $\models \forall x \in S.p$ means for **all** state σ , it is the case that $\sigma \models \forall x \in S.p$.
- (g) $\not\models \exists x \in S.p$ means for **some** state σ , it is the case that $\sigma \not\models \exists x \in S.p$.
- (h) $\not\models \forall x \in S.p$ means for **some** state σ , it is the case that $\sigma \not\models \forall x \in S.p$.